

NACOTE CREEK, NJ, USA

WMO: 998002

Lat: 39.5347N

Lon: 74.4636W

Elev: 12

StdP: 101.18

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-8.9	-20.1	0.6	-9.2	-17.3	0.8	-6.1	8.5	6.7	8.0	7.2	2.7	250	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	32.8	23.8	31.1	23.1	29.6	22.7	25.3	30.0	24.7	28.9	24.0	27.6	3.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.1	19.0	27.3	23.5	18.4	26.8	22.9	17.6	26.3	77.5	30.1	74.8	28.8	72.1	27.6	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.5	6.7	-13.1	34.8	3.2	1.5	-15.5	35.9	-17.4	36.7	-19.2	37.6	-21.5	38.6		
(5)			-14.0	26.6	3.0	0.4	-16.1	26.9	-17.9	27.1	-19.6	27.3	-21.7	27.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	-0.2	1.8	5.2	11.5	17.0	21.2	23.4	21.9	19.3	13.9	7.9	4.6		
	DBStd	9.08	5.04	4.94	4.59	4.37	4.16	3.69	3.51	4.16	4.06	4.08	4.41	4.81		
	HDD10.0	1029	318	234	161	29	1	0	0	1	1	12	91	181		
	HDD18.3	2697	575	463	408	213	77	13	7	20	36	146	313	426		
	CDD10.0	1884	1	4	12	74	217	337	416	369	279	133	28	13		
	CDD18.3	509	0	0	0	7	35	100	164	130	64	9	0	0		
	CDH23.3	3831	0	0	1	71	259	773	1366	954	361	46	0	0		
	CDH26.7	1208	0	0	0	24	66	281	476	266	86	9	0	0		
(14) Wind	WSAvg	2.9	2.9	3.1	3.4	3.6	3.2	2.8	2.6	2.3	2.7	2.8	2.7	2.7		
(15) Precipitation	PrecAvg	1143	97	73	114	87	86	83	103	120	94	93	83	110		
	PrecMax	1483	200	148	211	163	213	173	196	223	167	224	168	224		
	PrecMin	862	36	17	12	38	34	30	32	43	19	16	25	44		
	PrecStd	137	39	32	51	33	41	38	38	51	47	53	41	50		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.5	19.1	21.9	30.1	30.8	34.8	34.8	32.5	32.1	27.5	21.0	19.3		
		MCWB	11.3	13.8	14.9	17.5	21.6	24.3	24.6	23.0	23.2	19.8	15.3	17.2		
	2%	DB	12.7	14.3	17.6	24.7	28.7	32.6	33.2	31.1	29.0	24.3	18.8	16.4		
		MCWB	10.7	10.7	11.8	16.2	21.2	23.8	24.4	23.1	22.7	18.5	14.9	14.9		
	5%	DB	10.5	11.5	14.5	21.0	26.6	30.0	31.2	29.7	27.0	22.0	16.6	14.4		
		MCWB	8.8	8.6	10.5	14.9	20.1	23.0	23.6	22.7	21.2	16.8	14.0	13.1		
	10%	DB	8.0	9.1	12.5	18.4	24.0	27.8	29.4	28.1	25.2	20.5	15.1	12.5		
		MCWB	5.6	6.6	9.6	13.2	18.6	21.3	23.2	21.8	21.0	17.3	13.1	10.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.4	14.5	15.6	19.5	23.3	25.6	26.5	25.8	25.1	22.0	18.5	17.6		
		MCDB	14.1	17.7	19.3	25.7	29.0	32.1	32.6	29.5	29.4	25.0	19.8	18.8		
	2%	WB	11.1	11.6	13.3	17.7	22.2	24.6	25.6	25.0	24.1	20.3	15.8	15.1		
		MCDB	12.0	13.3	15.7	22.3	27.0	30.5	30.7	28.4	26.9	22.2	16.9	16.2		
	5%	WB	8.9	9.1	11.6	16.3	21.2	23.7	24.9	24.3	23.0	19.3	14.8	13.4		
		MCDB	10.6	10.9	13.8	20.5	25.2	28.6	29.3	27.1	25.6	21.1	16.1	14.4		
	10%	WB	5.9	6.8	10.0	14.7	19.9	22.7	24.2	23.6	22.2	17.9	13.3	11.3		
		MCDB	7.6	9.0	12.6	17.3	22.8	26.5	27.8	26.4	24.5	19.8	15.0	12.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.4	9.2	10.2	9.5	9.3	9.6	9.6	9.1	9.7	9.7	8.9		
		MCDBR	12.0	12.7	12.9	13.9	12.6	11.9	11.4	10.7	10.6	11.3	12.0	9.9		
	5% WB	MCWBR	10.5	9.4	8.7	7.0	6.2	4.7	4.4	4.4	5.1	6.7	8.6	8.5		
		MCWBR	11.7	11.9	11.0	11.9	10.7	11.3	9.9	8.5	8.7	8.8	9.3	9.3		
(40) Clear-Sky Solar Irradiance	taub	taub	0.316	0.321	0.355	0.387	0.433	0.490	0.497	0.490	0.411	0.363	0.332	0.315		
		taud	2.491	2.478	2.422	2.355	2.262	2.141	2.132	2.144	2.340	2.477	2.517	2.535		
	Edn at Noon	Edn at Noon	855	902	901	888	851	800	789	784	831	842	829	830		
		Edh at Noon	78	90	106	120	135	152	153	147	114	89	76	70		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.99	2.91	3.90	5.00	5.55	6.08	6.20	5.43	4.45	3.19	2.28	1.77		
	RadStd	RadStd	0.17	0.32	0.32	0.39	0.66	0.51	0.30	0.37	0.41	0.31	0.21	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+123	+72

Nomenclature: See separate page